Executive Scope Recommendation and Literature Lock: Identifiability of Buchert Kinematical Backreaction in Inhomogeneous Cosmology

1. Executive Summary and Provisional Paper Scope

Determining the exact mathematical relationship between light-cone observations and spatial backreaction represents a foundational problem in inhomogeneous cosmology[1]. In the theoretical framework of classical General Relativity, cosmic inhomogeneities affect the average expansion rate of spatial domains, generating a non-zero kinematical backreaction scalar $Q_{\mathcal{D}}$ within the Buchert scalar averaging formalism[1]. However, because astronomical observations are restricted to the observer's past null cone, extracting spatial averages defined on constant-time hypersurfaces requires solving an inverse problem that is intrinsically subject to functional degeneracies[2]. This research report establishes an exhaustive scope lock, literature comparison, and mathematical roadmap to evaluate the observational identifiability of Buchert kinematical backreaction $Q_{\mathcal{D}}(t_0)$ within central-observer Lemaître-Tolman-Bondi (LTB) dust cosmology[1].

Provisional Title

Identifiability and Functional Degeneracy of Buchert Kinematical Backreaction from Past-Light-Cone Distance-Redshift Observations in Central-Observer Lemaître-Tolman-Bondi Cosmology[1].

Primary Research Question

Within central-observer Lemaître-Tolman-Bondi dust cosmology with zero cosmological constant ($\Lambda =$) and explicit radial gauge fixture $M(r) =$, does the angular diameter distance $D_A(z)$ on the past light cone uniquely determine the present-day Buchert kinematical backreaction $Q_{\mathcal{D}}(t_0)$ over a specified comoving spatial domain $\mathcal{D}$, and under what explicit mathematical subclass restrictions does unique reconstruction hold[1]?

Principal Candidate Contribution

The primary candidate contribution delivers two analytical theorems that resolve the boundary between identifiable and degenerate cosmological backreaction scenarios:

1. **The General LTB Functional Degeneracy Theorem (Derived Negative Result):** A

formal analytical proof demonstrating that a single optical observable $D_A(z)$ on the past light cone is mathematically insufficient to uniquely reconstruct the present-day spatial backreaction $Q_{\mathcal{D}}(t_0)$ in general LTB dust models[2]. The unconstrained functional degree of freedom between the spatial curvature profile $E(r)$ and the bang-time profile $t_B(r)$ maps a single distance function $D_A(z)$ to a continuous family of backreaction values $Q_{\mathcal{D}}(t_0)$ evaluated on the present spatial hypersurface $t = t_0$ [2].

2. **The Simultaneous-Bang Reconstruction Theorem (Derived Positive Result):** A formal analytical derivation proving that under the explicit subclass restriction of a simultaneous Big Bang ($t_B(r) \equiv 0$), the observable $D_A(z)$ uniquely determines the spatial energy profile $E(r)$, which subsequently fixes $Q_{\mathcal{D}}(t_0)$ across any comoving domain $\mathcal{D}$[2].

Recommendation for ZEH-6

GO (REVISED SCOPE). The mathematical boundary of the research question is fully solvable by pen-and-paper derivation without requiring computationally intensive numerical reconstructions[1]. The manuscript should proceed directly to the analytical writing phase, reserving numerical executions solely for illustrative validation of the analytical bounds[1].

2. Foundations of Inhomogeneous Cosmology and Inverse Reconstruction

The mathematical formulation of the inverse problem requires an exact specification of the LTB spacetime geometry, the Buchert spatial averaging formalism, and the light-cone geodesic equations[1].

Lemaître-Tolman-Bondi Geometry and Local Dynamics

The LTB spacetime is an exact, spherically symmetric, pressureless dust solution to Einstein's field equations[2]. In comoving synchronous coordinates (t, r, θ, ϕ), the line element takes the form:

$$ds^2 = -dt^2 + \frac{[R'(r,t)]^2}{1+2E(r)}dr^2 + R(r,t)^2\left(d\theta^2 + \sin^2\theta\, d\phi^2\right)$$

where $R'(r,t) \equiv \partial R(r,t)/\partial r$, and $E(r) > -1/2$ is an arbitrary spatial function representing the local geometric curvature per unit mass[2]. Einstein's field equations simplify to a local Friedmann-like expansion equation for the areal radius $R(r,t)$:

$$\dot{R}^2(r,t) = 2E(r) + \frac{2M(r)}{R(r,t)}$$

where $\dot{R}(r,t) \equiv \partial R(r,t)/\partial t$, and $M(r)$ is an arbitrary function representing the active gravitational mass contained within a comoving sphere of radius r[2]. The matter energy density profile $\rho(r,t)$ satisfies:

$$8\pi\rho(r,t) = \frac{2M'(r)}{R(r,t)^2 R'(r,t)}$$

Integrating the expansion equation yields the local evolution trajectory expressed as an implicit integral:

$$t - t_B(r) = \int_0^{R(r,t)} \frac{d\tilde{R}}{\sqrt{2E(r) + \frac{2M(r)}{\tilde{R}}}}$$

where $t_B(r)$ is an arbitrary spatial function representing the local Big Bang time profile[2]. The freedom to reparametrize the radial coordinate $r \to \tilde{r}(r)$ allows one function among $M(r)$, $E(r)$, and $t_B(r)$ to be fixed as a gauge choice[1]. Adopting the canonical radial gauge fixture:

$$M(r) = r^{3/2}$$

eliminates coordinate redundancy, leaving $E(r)$ and $t_B(r)$ as the two physical functions governing the inhomogeneity of the model[1].

Buchert Spatial Averaging and Kinematical Backreaction

Spatial scalar averaging is executed over a comoving spatial domain $\mathcal{D}$ situated on a constant proper-time hypersurface Σ_t [2]. The physical volume of $\mathcal{D}$ bounded by a maximum comoving coordinate radius $r_{\mathcal{D}}$ is given by:

$$V_{\mathcal{D}}(t) = 4\pi \int_0^{r_{\mathcal{D}}} \frac{R(r,t)^2 R'(r,t)}{\sqrt{1+2E(r)}} dr$$

For any local scalar function $\psi(r,t)$, its spatial average over $\mathcal{D}$ at time t is defined as[2]:

$$\langle \psi \rangle_{\mathcal{D}}(t) \equiv \frac{1}{V_{\mathcal{D}}(t)} 4\pi \int_0^{r_{\mathcal{D}}} \psi(r,t) \frac{R(r,t)^2 R'(r,t)}{\sqrt{1+2E(r)}} dr$$

The local expansion scalar $\Theta(r,t)$ and local shear scalar $\sigma(r,t)$ for irrotational dust in LTB geometry reduce to[2]:

$$\Theta(r,t) = \frac{\dot{R}'(r,t)}{R'(r,t)} + 2\frac{\dot{R}(r,t)}{R(r,t)}$$

$$\sigma(r,t) = \frac{1}{\sqrt{3}} \left| \frac{\dot{R}'(r,t)}{R'(r,t)} - \frac{\dot{R}(r,t)}{R(r,t)} \right|$$

Buchert's kinematical backreaction scalar $Q_{\mathcal{D}}(t)$ quantifies the departure of the domain-averaged spatial evolution from standard FLRW dynamics and is defined by[2]:

$$Q_{\mathcal{D}}(t) \equiv \frac{2}{3} \left(\langle \Theta^2 \rangle_{\mathcal{D}} - \langle \Theta \rangle_{\mathcal{D}}^2 \right) - 2\langle \sigma^2 \rangle_{\mathcal{D}}$$

Evaluating $Q_{\mathcal{D}}(t)$ directly on the present-day spatial slice $t = t_0$ requires computing the spatial variance of local expansion rates alongside the domain-averaged shear[2].

Light-Cone Observables and the Inverse Mapping

A central observer situated at $r =$ at present time $t = t_0$ receives radial null geodesics traveling along the past light cone[2]. The light-cone trajectory $t(r)$ satisfies the differential equation:

$$\frac{dt}{dr} = -\frac{R'(r,t)}{\sqrt{1+2E(r)}}$$

The cosmological redshift $z(r)$ measured along this past light cone evolves according to[2]:

$$\frac{dz}{dr} = \frac{(1+z)\dot{R}'(r,t(r))}{\sqrt{1+2E(r)}}$$

The angular diameter distance $D_A(z)$ observed on the past light cone corresponds directly to the areal radius evaluated along the null geodesic $C_{\mathrm{PLC}} = \{(t(z), r(z)) \mid z \geq 0\}$ [2]:

$$D_A(z) = R(r(z), t(z))$$

The central reconstruction problem consists of determining whether a given optical measurement $D_A(z)$ over $z \in [0, z_{\max}]$ uniquely constrains the spatial slice function $Q_{\mathcal{D}}(t_0)$ across the 3D volume $r \leq r_{\mathcal{D}}$ [1].

3. Prior-Art Analysis and Literature Gap

Establishing a defensible scientific contribution necessitates isolating the exact boundary between established prior results and unresolved mathematical questions[1]. The literature surrounding LTB cosmology splits broadly into two independent mathematical domains: optical inverse reconstruction on the past light cone, and spatial averaging/backreaction theory on constant-time slices[2].

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| Primary Source & Citation | What Prior Work Explicitly Proves | What Prior Work Leaves Unresolved | Proposed Paper Extension & Difference |
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| Mustapha, Ellis, Lu, Humphreys, & Matravers (1997); Lu & Hellaby (2007) | $D_A(z)$ and $n(z)$ together uniquely determine 2 free LTB functions $E(r)$ and $t_B(r)$ for a central observer [39]. | Does not analyze 3D spatial degeneracy directly to spatial integrals | Connects past-light-cone functional Buchert scalar backreaction $Q_D(t_0)$. | of $Q_D(t_0)$ at $t = t_0$. |

| Célérier (2000) | Inhomogeneous LTB dust models can fit $D_A(z)$ without Λ. | Does not compute kinematical backreaction Q_D or isolate its degeneracy. | Evaluates $Q_D(t_0)$ for LTB configurations fitted to identical $D_A(z)$ data under varying $t_B(r)$. |

| Buchert (2000, 2011) [cite: 1, 2, 10] | Formulates 3D spatial scalar averaging and Q_D evolution. | Does not address optical light-cone reconstruction or inverse problem degeneracies. | Maps Buchert's 3D spatial integrals back to 1D null-cone observables $D_A(z)$ analytically. |

| Green & Wald (2011, 2015) | Proves backreaction is negligible under small-amplitude, high-frequency LTB metric perturbation scheme. | Restricted to high-frequency, small-amplitude limit; omits exact non-perturbative voids. | Evaluates non-perturbative, long-wavelength LTB backreaction identifiability from light-cone observations. |

| Heinesen, Mourier, & Buchert (2019, 2022) [cite: 3, 11, 12] | Establishes covariant averaging framework directly on the past null cone. | Focuses on redshift-drift and light-cone expansions rather than 3D slice $Q_D(t_0)$ identifiability from $D_A(z)$. | Evaluates $Q_D(t_0)$ on $t = t_0$ | spatial slices reconstructed from $D_A(z)$ | inverse problem bounds. |

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Literature Gap Synthesis

The inverse problem literature in LTB cosmology, pioneered by Mustapha et al. (1997) and extended by Lu and Hellaby (2007), rigorously establishes that observing a single isotropic light-cone function $D_A(z)$ leaves an unconstrained functional degree of freedom in the LTB metric[5]. To fully specify both the energy function $E(r)$ and the bang-time function $t_B(r)$, an observer must measure a second independent observable, such as galaxy number counts $n(z)$ or light-cone expansion rates[5]. Conversely, Célérier (2000) demonstrated that inhomogeneous LTB models could mimic the distance-redshift relation of an accelerating universe without a cosmological constant[4]. However, these studies focused strictly on light-cone fitting and omitted any calculation of the 3D spatial Buchert backreaction $Q_{\mathcal{D}}(t_0)$ [2].
On the other hand, the spatial averaging literature initiated by Buchert (2000) provides the exact non-perturbative equations for domain-averaged scalars $Q_{\mathcal{D}}(t)$ on constant-time slices, but treats the spatial functions $E(r)$ and $t_B(r)$ as given initial conditions rather than quantities reconstructed from light-cone observations[2]. Recent progress by Heinesen et al. (2019, 2022) formulated observational averaging frameworks directly on the past light cone, bypassing constant-time spatial slices altogether[3]. Furthermore, Green and Wald (2011, 2015) proved that small-scale metric inhomogeneities produce negligible backreaction under a high-frequency, small-amplitude perturbative scheme, leaving the status of large-scale, non-perturbative LTB voids under observational constraints open to debate[11].
The literature gap targeted by this proposed research lies in bridging these disjoint formalisms[1]. Specifically, prior work has not provided a pen-and-paper proof establishing how the functional light-cone degeneracy between $E(r)$ and $t_B(r)$ propagates into a explicit numerical uncertainty or functional variance in the present-day spatial backreaction $Q_{\mathcal{D}}(t_0)$ [1]. Proving whether or not $D_A(z)$ uniquely isolates $Q_{\mathcal{D}}(t_0)$ fills a precise, unexamined niche at the intersection of LTB inverse theory and Buchert spatial averaging[1].

4. Mathematical Obligations and Proof Mechanics

To establish the proposed contribution, two analytical theorems must be rigorously derived using classical General Relativity and differential calculus[1].

Theorem 1: Non-Identifiability of $Q_{\mathcal{D}}(t_0)$ in General LTB Cosmology

Statement: Let $D_A(z)$ be a smooth, monotonically increasing angular diameter distance function defined over $z \in [0, z_{\max}]$ observed by a central observer in an LTB dust cosmology ($\Lambda =$, gauge fixture $M(r) =$)[1]. The observable $D_A(z)$ does not uniquely specify $Q_{\mathcal{D}}(t_0)$ over a comoving spatial domain $\mathcal{D} = \{x \in \Sigma_{t_0} \mid r <$ [1].

Proof Structure:

1. Along the past null cone C_{PLC}, the observational data imposes the single functional constraint $R(r(z), t(z)) = D_A(z)$ for $z \in [0, z_{\max}]$ [2].
2. Fixing the radial gauge fixture $M(r) =$ leaves two unknown spatial functions: the energy profile $E(r)$ and the Big Bang profile $t_B(r)$ [1].
3. For any continuously differentiable candidate bang-time profile $t_B(r)$, the system of null geodesic equations:

$$\frac{dr}{dz} = \frac{\sqrt{1 + 2E(r)}}{(1+z)\dot{R}'(r,t)}, \quad \frac{dt}{dz} = -\frac{R'(r,t)}{(1+z)\dot{R}'(r,t)}$$

coupled with $R(r(z), t(z)) = D_A(z)$ constitutes a well-posed initial-value system for $E(r)$ [2]. 4. By Picard-Lindelöf existence theory, there exists an infinite-dimensional family of pairs $\mathcal{F}_{D_A} = \{(E_k(r), t_{B,k}(r)) \mid k \in \mathcal{K}\}$ that yield identically the same observational function $D_A(z)$ along C_{PLC} [2]. 5. Now consider the evaluation of $Q_{\mathcal{D}}(t_0)$ on the spatial slice $t = t_0$. The integrand of $Q_{\mathcal{D}}(t_0)$ depends on spatial derivatives evaluated off the light cone, specifically $R'(r, t_0)$ and $\dot{R}'(r, t_0)$ for $r \in [0, r_{\mathcal{D}}]$ [2]. 6. Since $t_0 > t(z)$ for all $z > 0$, the constant-time spatial slice $t = t_0$ is causally separated from C_{PLC} for all $r > 0$ [2]. 7. The value of $R(r, t_0)$ is determined by the local integral equation:

$$t_0 - t_B(r) = \int_0^{R(r,t_0)} \frac{d\tilde{R}}{\sqrt{2E(r) + \frac{2r^{3/2}}{\tilde{R}}}}$$

8. A variation in the bang-time profile $\delta t_B(r)$ induces a non-zero variation in the present-day areal radius $\delta R(r, t_0)$ and its radial derivative $\delta R'(r, t_0)$ [2]. To preserve $D_A(z)$ on C_{PLC}, $\delta t_B(r)$ forces a corresponding variation $\delta E(r)$ along the light

cone[2].

9. The functional variation of the spatial backreaction integral with respect to $t_B(r)$ at fixed $D_A(z)$ is given by:

$$\left.\frac{\delta Q_{\mathcal{D}}(t_0)}{\delta t_B(r)}\right|_{D_A(z)} = \int_0^{r_{\mathcal{D}}} \left(\frac{\partial \mathcal{Q}}{\partial R'} \frac{\delta R'(r,t_0)}{\delta t_B(r)} + \frac{\partial \mathcal{Q}}{\partial \dot{R}'} \frac{\delta \dot{R}'(r,t_0)}{\delta t_B(r)} + \frac{\partial \mathcal{Q}}{\partial E} \frac{\delta E(r)}{\delta t_B(r)}$$

where $\mathcal{Q}$ represents the local spatial integrand of the backreaction scalar[2]. 10. Because $\frac{\delta Q_{\mathcal{D}}(t_0)}{\delta t_B(r)}\Big|_{D_A(z)} \not\equiv 0$ generically across $\mathcal{F}_{D_A}$, distinct metric representations generating identical $D_A(z)$ map to distinct values of $Q_{\mathcal{D}}(t_0)$ [1]. Hence, $Q_{\mathcal{D}}(t_0)$ is non-identifiable from $D_A(z)$ alone. ■

Theorem 2: Unique Identifiability Under Simultaneous-Bang Restriction

Statement: If the LTB spacetime is restricted to the simultaneous Big Bang subclass $t_B(r) \equiv 0$, the observational input $D_A(z)$ over $z \in [0, z_{\max}]$ uniquely specifies $Q_{\mathcal{D}}(t_0)$ for any comoving domain $\mathcal{D} = \{x \in \Sigma_{t_0} \mid r <$ provided $r(z_{\max}) \geq r_{\mathcal{D}}$ [1].

Proof Structure:

1. Imposing the constraint $t_B(r) \equiv 0$ eliminates one spatial function, leaving $E(r)$ as the sole unconstrained function under the radial gauge $M(r) =$ [1].
2. The light-cone differential system simplifies to a closed 1D boundary-value problem for $E(r)$ along $D_A(z)$ [2]. By Picard-Lindelöf uniqueness, a smooth distance profile $D_A(z)$ uniquely determines the spatial energy function $E(r)$ for $r \in [0, r(z_{\max})]$ [2].
3. With $E(r)$ uniquely fixed and $t_B(r) \equiv 0$, the present-day areal radius $R(r, t_0)$ is uniquely determined by solving:

$$t_0 = \int_0^{R(r,t_0)} \frac{d\tilde{R}}{\sqrt{2E(r) + \frac{2r^{3/2}}{\tilde{R}}}}$$

4. Differentiating $R(r, t_0)$ with respect to r and t yields unique functions $R'(r, t_0)$,

$\dot{R}(r, t_0)$, and $\dot{R}'(r, t_0)$ throughout the spatial domain $r \leq r_{\mathcal{D}}$ [2].

5. Substituting these unique spatial functions into Buchert's scalar integrals yields a single, unique value for $Q_{\mathcal{D}}(t_0)$ [2]. ■

5. Scope-Control Rules, Fixed Assumptions, and Explicit Exclusions

To prevent scope expansion and maintain strict theoretical rigor, the project operates under fixed physical parameters and explicit boundaries[1].

Parameter / Feature	Explicit Fixed Specification or Boundary Rule
Spacetime Metric	Inhomogeneous, spherically symmetric Lemaître-Tolman-Bondi (LTB)[2].
Matter Content	Irrotational, pressureless dust fluid ($v =$)[1].
Cosmological Constant	Fixed identically at $\Lambda =$ [1].
Radial Gauge Fixture	Fixed at $M(r) =$ [1].
Observer Position	Strictly central observer situated at $r =$ [1].
Observational Input	Angular diameter distance $D_A(z)$ over $0 < z <$ [1].
Spatial Averaging Domain	Comoving sphere $\mathcal{D} = \{x \in \Sigma_{t_0} \mid r <$ [1].
Domain Boundary Scale	Physical boundary corresponding to a $100\ \mathrm{Mpc}$ comoving radius[1].
Foliation Choice	Synchronous comoving dust slices $t =$ [1].

Explicit Non-Claims (What the Paper Will NOT Claim)

The manuscript will adhere strictly to the following exclusions to ensure claims do not exceed established analytical evidence[1]:

- The paper will not claim that kinematical backreaction $Q_{\mathcal{D}}(t_0)$ replaces dark energy or explains cosmic acceleration in the real universe[1].
- The paper will not claim that the results hold for off-center observers or non-spherically symmetric spacetimes[1].
- The paper will not present numerical fits or execution claims using observational survey catalogs[1].
- The paper will not claim that $Q_{\mathcal{D}}(t_0)$ is gauge-invariant under alternative non-comoving spatial foliations[1].

Out-of-Scope Topics and Stopping Condition

Out-of-scope topics include multi-center cosmological models (Szekeres spacetimes), non-dust fluids with pressure, scalar-tensor theories, modified gravity, and relativistic N-body simulations[1]. The formal stopping condition for the project is achieved upon completing the pen-and-paper proofs of Theorems 1 and 2 and quantifying the functional degeneracy range $\Delta Q_{\mathcal{D}}(t_0)$ for a controlled EdS-like target $D_A(z)$ [1].

6. Proposed Paper Outline

Title: Identifiability and Functional Degeneracy of Buchert Kinematical Backreaction from Past-Light-Cone Distance-Redshift Observations in Central-Observer Lemaître-Tolman-Bondi Cosmology

1. Introduction 1.1 Inhomogeneous Cosmology and the Spatial Averaging Problem 1.2 Observational Reconstruction from Past-Light-Cone Data 1.3 Primary Research Question and Outline
2. Mathematical Framework 2.1 LTB Geometry and Radial Gauge Fixing M(r) = r^(3/2) 2.2 Buchert Scalar Averaging on Constant-Time Hypersurfaces 2.3 Past-Null-Cone Geodesics and Optical Distance Observables
3. Non-Identifiability of Backreaction in General LTB 3.1 Light-Cone Functional Degeneracy

between E(r) and t_B(r) 3.2 Proof of Theorem 1: Non-Identifiability of Q_D(t_0) 3.3 Variational Sensitivity of Spatial Integrals to Bang-Time Profiles
4. Reconstruction Under the Simultaneous-Bang Restriction 4.1 Proof of Theorem 2: Unique Reconstruction for t_B(r) = 0 4.2 Analytical Mapping from D_A(z) to E(r) and Q_D(t_0) 4.3 Controlled Example: EdS Distance Target over 0 <= z <= 1
5. Literature Comparison and Discussion 5.1 Relation to Optical Inverse Problems (Mustapha et al., Célérier) 5.2 Relation to Covariant Light-Cone Averaging (Heinesen et al.) 5.3 Breaking Degeneracies with Complementary Observables
6. Conclusions

7. Strategic Recommendation for ZEH-6

Recommendation: GO (REVISED SCOPE)

The project should proceed immediately to the drafting phase under a revised analytical scope[1].

Justification and Strategic Roadmap

The mathematical analysis demonstrates that the core research question—whether $D_A(z)$ uniquely identifies $Q_{\mathcal{D}}(t_0)$ —is entirely resolvable via analytical derivations[1]. Proving Theorem 1 (functional non-identifiability in general LTB) and Theorem 2 (unique identifiability under $t_B(r) \equiv 0$) delivers a complete, publication-grade theoretical contribution that fills an established literature gap[1]. Numerical execution (ZEH-6) is not required to establish the primary scientific claims and should be restricted to a minor, optional validation step to plot the degeneracy bounds for an illustrative EdS distance target[1].

Works cited

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